Evidence-Based Fake News Detection with Calibrated Scoring

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Abstract. Most automated fake news detectors score an article on how it is written, and all of that is chosen by the author, so a fabricated story in a calm newsroom style can score as well as a genuine report. We built a checking system that weighs evidence outside the author's control instead. It accepts links, pasted text, screenshots and videos, reads what each message asserts in English, Tamil or Hindi, whether stated, asked or passed on, and refuses input with no checkable claim. Each claim is checked against independent news coverage, counting outlets rather than articles, against rulings already published by fact-checkers, against the current week's coverage of its subject, and against an encyclopedia for false premises, and the publisher is identified from its web address. Writing signals become likelihood ratios estimated from labelled articles and added as log-odds, but may only count against a story. The output is a calibrated probability and a verdict with its evidence, or an abstention. On 120 held-out ISOT articles the system matched a hand-weighted baseline to within one article and lowered calibration error on its accusations from 0.267 to 0.152, and on six careful fabrications the baseline rated four credible and ours none. On 22 claims about the current month it answered 22 correctly, against 15 for the same language model asked without evidence. On 200 claims ruled on by fact-checkers and sealed before development, it gave the fact-checker's or an acceptable verdict for 141 and a wrong verdict for 12.

Keywords: Fake news detection, Evidence-based verification, Independent corroboration, Likelihood ratio, Calibration, Abstention.

1 Introduction

In its *Global Risks Report 2026*, the World Economic Forum ranked misinformation and disinformation as the second most severe global risk over the next two years, just behind

geoeconomic confrontation [2]. Language models now make convincing fakes cheap to produce [3]. Most people meet them as a forwarded message, a screenshot or a link, with no easy way to check where they came from.

Automated detectors are supposed to help with that. Most of them treat the problem as text classification: a model reads the article and labels it fake or real from its wording. Scores on public datasets are high. On the ISOT corpus, accuracies above 99% are common, but an audit by Verma [4] traced much of that to the way the corpus was put together. There is a simpler problem too. A style-based detector rewards features that the writer controls. Anyone fabricating a story can keep the headline calm, quote named officials and avoid loaded words. Chen and Shu [3] found that misinformation rewritten by a language model is harder for both people and detectors to catch than the human-written version with the same meaning.

We saw this in our own project. Our first design added up six checks with hand-picked weights. When we tested it with stories we had invented and written in plain newsroom style, they averaged 7.3 out of 10, and four of the six deceptive items came out as credible. Each check was doing its job. The sum was the problem, because good writing could make up for missing evidence.

This paper describes what we built instead. One rule drives it: anything the author controls may count against a story but never in its favour. Support has to come from outside, from independent outlets reporting the same event, from reference entries on the people named, and from the publisher’s record, which we take from the web address. Each observation contributes a likelihood ratio estimated from labelled articles. The system reports a probability, a verdict and the rule that produced the verdict, and when a check cannot run it says so instead of treating the gap as evidence.

Our base paper is DEFAME [1], a fact-checking system published at ICML 2025 that plans web searches, gathers text and image evidence and writes a justified verdict. We keep its evidence-first approach and add four things a public tool needs: limits on what each kind of evidence may prove, a count of independent sources, a check on who published the page, and defined behaviour when a check fails. We also measured which language-model tasks can run on a local model, and nine of ten can.

Section 2 reviews related work. Section 3 describes the system and the evaluation. Section 4 reports the results, Sect. 5 discusses them, and Sect. 6 concludes.

2 Related Work

Content-based detectors classify the article text. Amri et al. [5] built ExFake, which combines the article with its social context and outside evidence to explain each decision. Its strongest signals, such as who shares a post, only arrive after a story has spread, and an article that has just been forwarded to someone has none. Da San Martino et al. [6] defined a span-level taxonomy of persuasion techniques for SemEval-2020, which we use for one of our writing checks.

Large language models have been tried both as detectors and as helpers. Hu et al. [7] found that GPT-3.5 used directly did worse than a fine-tuned small model, but its written rationales helped the small model when given as extra input. Wang et al. [8] had a language model argue both sides of a claim from collections of raw reports and then pick the better-supported explanation. Both still judge the content or rely on many reports per claim.

Evidence-based fact-checking tests claims against outside sources. Guo et al. [9] describe the whole pipeline, from claim detection through evidence retrieval and verdict prediction to justification, and point out that most systems assumed the evidence was already at hand. AVeriTeC [10] pairs real-world claims with evidence from the open web.

DEFAME [1] is the closest system to ours. It runs in six stages (plan, execute, summarise, develop the fact-check, predict, justify), uses web search, image search, reverse image search and a geolocation tool, and returns supported, refuted, misleading or not enough information. If the evidence is insufficient, it goes back to planning, up to three times. With GPT-4o it reported 70.5% accuracy on AVeriTeC, 59.2% on MOCHEG, 83.9% on the true/false split of VERITE and 69.7% on ClaimReview2024+, at about 13 US cents and four minutes per claim. Its authors note that it does not weigh source credibility and relies on the search engine’s own trustworthiness checks. It also gives a label without a probability.

Source ratings are another way in. Lin et al. [11] found that six professional rating sets for news sites agree closely, and they six professional rating sets into one score for 11,520 news domains. A rating only helps if the outlet is identified correctly. A system that trusts the name printed on a page will give a copycat site the original’s record.

Calibration and abstention come from machine learning more broadly. Minderer et al. [12] found that calibration varies a lot from one model family to another, so a stated confidence has to be measured before it can be read as a probability. Silva Filho et al. [13] surveyed how to measure and repair calibration, including isotonic regression. Hendrickx et

al. [14] surveyed models that may decline to answer and how such refusals should be judged. Table 1 groups the cited work.

Table 1. Families of related work and how this paper relates to them.

Family	Studies	Evidence and output	Relation to this work
Content classifiers	[4, 5, 6]	Article text, social context; a label	Writing signals kept, but one-sided
LLM-assisted detection	[7, 8]	Text, rationales, raw reports; label and explanation	Model used only for narrow tasks
Evidence-based fact-checking	[1, 9, 10]	Wikipedia or web evidence; verdict and justification	Base approach; we add independence and identity checks
Source ratings	[11]	Ratings of outlets or domains	Used after the publisher is identified by address
Calibration and abstention	[12, 13, 14]	Probabilities, reject option	Calibrated probability; defined abstention

3 System and Method

3.1 Task

The input is anything a reader might want checked: a link to an article, pasted text, a screenshot or a video link. The output is one of six verdicts (REAL, FAKE, PARTLY FALSE, CANNOT VERIFY, NOT A FACTUAL CLAIM or SATIRE), a probability that the story is fabricated, and the evidence behind the verdict. Input with nothing checkable in it is refused with a short reason. We work with claims, not whole articles, because one story can mix true and false statements and because claims are what other outlets report. Fig. 1 shows the principle behind the scoring and Fig. 2 the pipeline.

3.2 Reading the Input

A link is read from the publisher's page. A news API returns only about 200 characters of an article, while the page gives 2,000 to 6,000. Page extraction took several fixes. BBC pages had 76 opening `<p>` tags and only 37 closing ones, and our first extractor read the site menu as the first paragraph of the story. Extraction is now limited to the article body, and the text is used only if at least half of the headline's distinctive words appear in it. Screenshots are read with Tesseract OCR and videos through their caption tracks. Text in Tamil, Hindi and Kannada is detected, and its claims are searched in the original language and in English, since a story reported in Tamil is often reported in English as well.

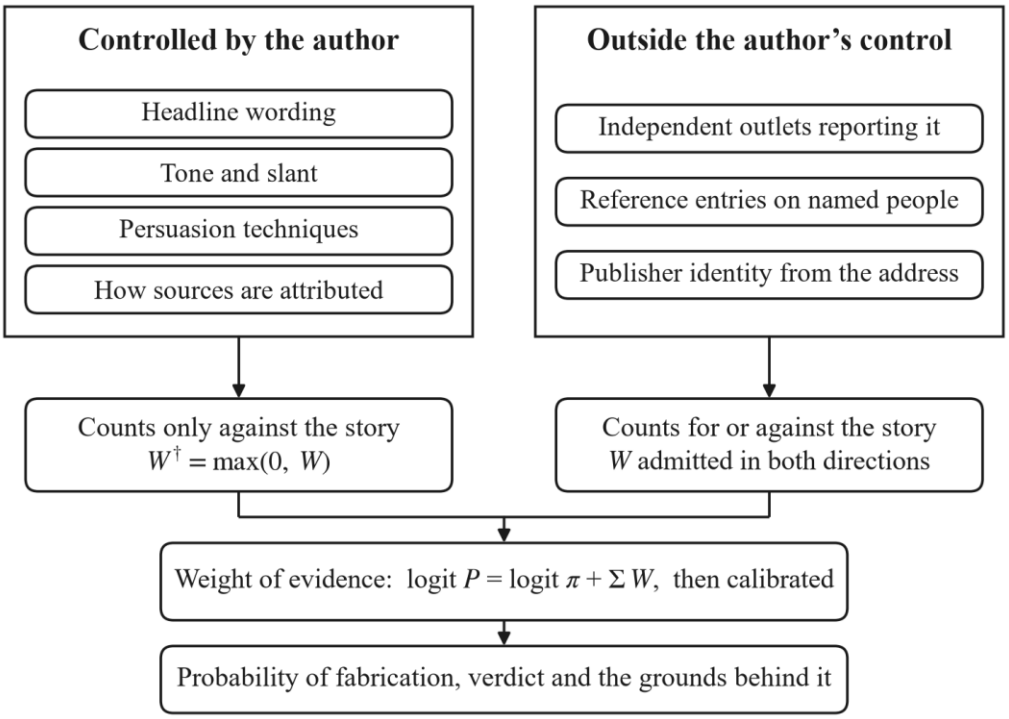


Fig. 1. The scoring principle. Observations the author controls can only count against a story; evidence from outside the author's control can count either way.

A pasted message or a screenshot is then read for what it asserts. People rarely paste a tidy claim: they ask ('is it true that RBI banned 500 rupee notes?'), pass something on ('I heard Modi resigned'), forward warnings in capitals, mock a claim sarcastically, or write in Tamil typed in English letters. One hosted-model call at temperature zero returns the language, the kind of message, its tone, an English rendering and up to three claims, each a neutral English sentence with its time frame, kind, subjects and search terms. A question is checked as the claim it asks about and answered directly, a hedge as the claim passed on, and sarcasm as the claim being mocked; negation is kept as stated, and a fact that an opinion merely takes for granted is not listed. Input with no checkable claim, such as a pure opinion, a forecast or a call to action, is refused with the reason.

Before this step existed, a rule-based check alone decided what could be checked, and it looked for capitalised names. It refused every question, and refused Tamil and Hindi text as naming no one, since those scripts have no capitals. It remains as the fallback when the reading step cannot run, so the page loses precision rather than failing.

3.3 Evidence from Outside the Article

Corroboration. For an article, a language model pulls up to two checkable claims out of the text, each with three or four search keywords; for a message, the claims come from the

reading step. Each claim is searched live in Google News, as a sentence, as a headline and by keyword, with NewsAPI as a fallback and the article's own outlet left out. When a claim states a figure, the competing figures reported in headlines are searched too, so '84 medals' meets the reports of 85. A second model call reads each headline and decides whether it concerns the same event, whether both could be true, and whether it supports or contradicts the claim, quoting the words that state the reported fact. The results are then reduced to the sources behind them: outlets with a common owner count once, copies of one wire-agency story count once, and near-duplicate reprints are merged. A relevance gate drops coverage that shares no anchor with the claim. We made the gate stricter after one of our invented test stories, about a chemical leak and written without a single proper noun, was 'confirmed' by a real chemical-leak report from Ohio that happened to share four ordinary words with it. A claim that names nothing can no longer be corroborated.

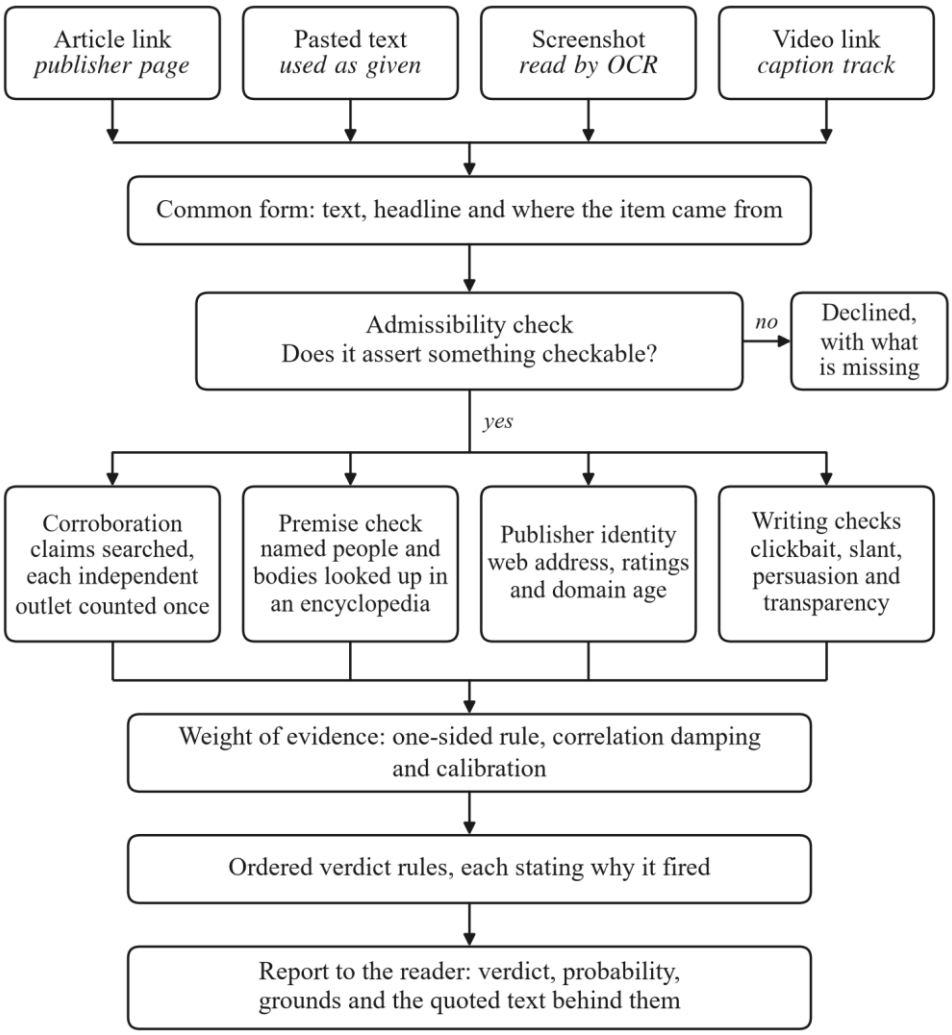


Fig. 2. Processing pipeline, from the reader's input to the report.

A search can fail for reasons that have nothing to do with the story: a rate limit, a timeout, a reply that cannot be parsed. Our first version treated all of these as ‘no coverage’ and told readers that no other outlet was reporting a wire story that five major outlets were carrying at the time. Each search now records whether it ran. A failed search is reported as unavailable and adds nothing to the probability. Our first index, NewsAPI’s free plan, ran a day behind live publishing: on 28 September 2026 the newest article it returned for any query was 30.5 hours old. Google News is searched first now, and only a story less than three hours old is reported as too recent to corroborate.

Premise check. A story built on a false premise describes something nobody reports, so a news search finds nothing and cannot say the premise is wrong. The system looks up the people and organisations a story names in Wikipedia and asks a model whether the story conflicts with the entry. A conflict counts only if the model quotes the entry and the quote is found in it, if the entry is about the same person, if the two statements cannot both be true, and if the story makes the claim itself instead of quoting someone else. Each condition came from a real error. One of our genuine test articles quoted a council official called Meera Krishnan and was flagged because the Wikipedia entry under that name is about an actress. An Associated Press video was flagged in one of eight runs because a woman interviewed in it said the Pope’s ancestors came from the region, which the check set against his birthplace in Chicago. The reference work may also confirm a standing fact, such as a state’s capital, in a quoted sentence, but never an event of the day.

Fact-checker rulings. Viral rumours are rarely reported as news, so the news search can only fail to find them. Desks such as AFP, PTI, India Today, The Quint, BOOM and Factly publish their rulings with ClaimReview markup that Google indexes. We search that index with short phrasings, since it matches every word, and Google News for fact-check headlines. A ruling on a similar but different claim must not count, so a model is shown every candidate and asked whether it examined the same assertion and what its finding makes of the user’s text: a ruling that rated ‘the video is genuine’ false makes ‘the video is fake’ true. A matched false ruling decides the verdict, with each desk, rating, date and link shown.

Current coverage. An invented event about a heavily covered subject leaves a trace in the rest of the week’s news. On the day a message said the Prime Minister had resigned, outlets reported him launching projects in Chennai. We read the last seven days of Google News coverage for each subject of the claim, and a model marks headlines that could not be true if the claim were, quoting the words that show it. Two independent outlets of this kind make a

claim probably false. A weaker form applies only to an event that would have dominated a subject covered every day and is absent from at least twelve headlines from five outlets. Both apply only when the news search found the event neither reported nor contradicted.

Publisher identity. The publisher is decided from the web address, not from the name the page gives itself. An outlet's record applies only on a domain it publishes from or on a platform channel listed as its own. Ratings come first from our list of 90 outlets, which includes Indian publishers missing from the published sets, and otherwise from the ratings of Lin et al. [11] for 11,520 domains. On the 81 outlets in both, the two agree well (Pearson r = 0.848, Spearman ρ = 0.765), and a least-squares fit over those 81 (reliability = 0.38 + 8.95 × score) maps the published 0–1 score onto our 0–10 scale. A page that uses a known outlet's name from an address the outlet does not use gets no record and is flagged as impersonation. For unrated sites the system reads the domain's registration date from RDAP. A domain under 90 days old counts as a warning sign, while an old domain earns nothing, because old domains can be bought.

3.4 Writing Checks

Four checks look at how the story is written: clickbait in the headline, slanted or one-sided language, persuasion techniques from the SemEval-2020 taxonomy [6], and transparency, a five-point checklist covering named sources, attributed quotes, links to primary evidence, concrete detail and measured claims. Each check has to quote the words it is reacting to, so its output can be compared with the text. The reader sees all four. How much they may affect the result is set by the scoring rule in Sect. 3.5.

3.5 Scoring

Each observation o is turned into a likelihood ratio and expressed in decibans:

$$\Lambda(o) = \frac{P(o \mid \text{fabricated})}{P(o \mid \text{genuine})}, \qquad W(o) = 10 \log_{10} \Lambda(o) \tag{1}$$

The ratios are estimated from labelled articles by putting each observation into bands and counting, with a Jeffreys prior. Each estimate is shrunk to the conservative end of its 95% interval, so a band with only a few articles behind it contributes nothing. The log-odds of fabrication are then a sum:

$$\operatorname{logit} P(\text{fabricated} \mid O) = \operatorname{logit} \pi + \sum_{o \in O} W(o) \tag{2}$$

where π is a prior set by the publisher's record. We use 0.02 for an outlet with a strong record, 0.06 for an ordinary one, 0.15 for an unrated outlet, 0.25 for a poor record or an impersonating page and 0.35 for a social-media post. These values are our own choice, not measurements, and every report says so.

Observations are split into O_{A}, which the author controls, and O_{I}, which the author does not. For $o \in O_{\mathrm{A}}$ only the part that counts against the story is kept:

$$W^{\dagger}(o) = \max(0, W(o)), \qquad o \in O_A \tag{3}$$

On a corpus of crude fakes, fluent writing really does predict a genuine article, and an unconstrained model learns to reward it. A careful fabricator writes fluently on purpose, so that lesson fails when it matters most. A side effect of Eq. (3) is that a check that tells us nothing, with a ratio of one, adds exactly zero, and so does a band that has never been measured. The report names such checks instead of giving them a made-up weight.

The four writing checks read the same prose and are correlated. Their mean pairwise correlation $\bar{r}$ was 0.55, and headline quality and slant correlated at 0.93. With k checks and equal correlation, the effective number of independent checks is

$$n_{\mathrm{eff}} = \frac{k}{1 + (k-1)\,\overline{r}} \tag{4}$$

which is 1.52 for our four, so their total weight is multiplied by $n_{\mathrm{eff}}/k = 0.379$ and capped at 8 decibans. Finally, the summed log-odds are calibrated with isotonic regression [13] on held-out data. When we first fitted the map, it quietly put back the weight Eq. (3) had removed, because it was fitted on ISOT, where clean style does go with genuine articles. The clamp is now applied again after calibration, and a regression test checks it.

3.6 Verdict Rules

The probability and the verdict answer different questions. The probability is graded belief. The verdict says what has been established. It comes from an ordered list of rules (Table 2), the first rule that matches decides, and its name is shown to the reader. The two can disagree, and that is deliberate. An uncorroborated story may get a low probability simply because most stories are genuine, but nothing about it has been confirmed, so it stays CANNOT VERIFY.

Table 2. Verdict rules in order of precedence. Input that fails the admissibility check is refused before these rules are reached.

Order	Condition	Verdict shown
1	The publisher is a known satirical outlet	SATIRE
2	Fact-checkers rated this same claim false or misleading	FAKE
3	One claim is confirmed and another contradicted	PARTLY FALSE
4	Two or more independent outlets contradict a claim	FAKE (false)
5	One independent outlet contradicts a claim	FAKE (probably false)
6	A reference entry contradicts the story's premise	FAKE (probably false)
7	Fact-checkers rated the claim true	REAL (probably true)
8	The story is too recent, or verification could not run	CANNOT VERIFY
9	No checkable claim is left	NOT A FACTUAL CLAIM
10	A claim made as new is confirmed only by coverage weeks old	FAKE (old news)
11	A confirmed claim carries one that nothing reports	PARTLY FALSE or CANNOT VERIFY
12	Two or more independent outlets support the claims	REAL (confirmed)
13	One independent outlet supports the claims	REAL (probably true)
14	A reference entry states a standing fact	REAL (probably true)
15	The week's coverage of the subject rules the claim out	FAKE (probably false)
16	A dominant event is absent from heavy coverage	FAKE (probably false)
17	A high-impact claim is uncorroborated and shows three or more risk signals	FAKE (probably false)
18	None of the above	CANNOT VERIFY

3.7 Model Routing

Every language-model call goes through one gateway with three providers: a local Llama 3.1 8B model served by Ollama, NVIDIA NIM and Google Gemini. We chose the provider for each task by measurement. A task goes to the local model only if at least 80% of its local replies are valid, at least 80% are correct where the answer is known, and its scores stay within 2 points of the hosted reference on the 0–10 scale. The gateway retries congested providers and moves on when a model is withdrawn. That happened to us once: NVIDIA retired `meta/llama-3.1-8b-instruct`, the model our first version used, on 26 August 2026, and every analysis began failing with HTTP 410. If no provider answers, each writing check falls back to word lists and surface patterns, and the report marks it as degraded.

Six judgements were later moved to Gemini by hand, each after a live failure that the fixed-input benchmark could not see: the local model read ‘TVK sweeps both seats’ as agreeing with ‘TVK lost’, and translated the Tamil தவெக (TVK) as DMK. Gemini’s free tier gives each model a daily quota, 500 requests for Flash-Lite, so the gateway tries six models in turn and skips one whose quota is spent until it resets, instead of retrying it, which had once stretched a check to 170 seconds. Judgements made at temperature zero are kept for thirty minutes, so a repeated check makes no model call. The premise check, which the local model failed, and the matching of a new claim against earlier checks are routed the same way. Our deployed server, a free instance with 512 MB of memory and no GPU, cannot run the local model, so there the measured tasks start at NIM.

The application is built with MongoDB, Express, React and Node.js. Reports are cached by article address for 14 days, each evidence channel streams its finding to the reader as it lands, and the code has 519 backend and 42 frontend tests that run without network access or API keys, rerun from a clean copy of the code on every change.

3.8 Evaluation

We used five test sets and one benchmark of the model providers.

Held-out ISOT articles. After length filtering, ISOT has 21,379 genuine and 21,915 fake articles. We used the 239 articles whose writing checks had all come from the language model and not from the fallback analysers. A hash of the article ID split them into 119 for estimating the ratios and 120 for testing. All of ISOT’s genuine articles come from Reuters, so we hid the source and stripped agency prefixes such as ‘WASHINGTON (Reuters)’, which would otherwise give the label away. Historical articles have no live coverage, so corroboration plays no part here, and this test covers only the writing half of the model. We had planned to use 300 articles but stopped at 239 when both hosted providers went down during the run, one with HTTP 503 errors and the other out of quota. The baseline is our earlier hand-weighted sum applied to the same check outputs. We report accuracy with 95% Wilson intervals, ROC-AUC, the Brier score and the expected calibration error (ECE) over ten bins. We also compute ECE and precision over accusations alone, meaning the predictions that put the probability of fabrication above 0.5.

Adversarial set. ISOT’s fakes are crude, so it cannot show whether good writing still buys trust. We wrote ten items: five fabrications in plain newsroom style with named sources and no clickbait, one high-impact fabrication with manipulation signals, two genuine reports, one

satirical piece and one opinion column. Each item was scored under both rules from the same check outputs, using the rule-based analysers so that the run can be repeated without network access.

End-to-end cases. Twelve cases went through the running application on 4 October 2026: pasted text, Tamil text, news links, screenshots and a YouTube video, with every verdict represented.

Messages as people write them. A robustness suite of 55 messages, each labelled by hand from October 2026 reporting, covers questions, hedges, forwards in capitals, sarcasm, negation, opinion around a fact, several claims at once, units and typos, Tamil, Tanglish and Hindi, old news, standing facts, quotes, local news, forecasts and vague text. A held-out set of 27 cases on new topics was written after development and run once before any change against it. We count an answer as expected, acceptable (a listed abstention), missed (an abstention where a verdict was due) or wrong (a verdict the wrong way).

Plain-model benchmark. 22 claims about October 2026, from wrong figures to true reports, were put both to the full system and to Gemini 3.1 Flash-Lite asked zero-shot with the date and nothing to read.

Fact-checked rulings. The sets above were written by us. This one takes its claims from others: 320 claims on which 29 fact-checking desks, among them FACTLY, NewsMobile, AFP Fact Check, The Quint, BOOM and Full Fact, published rulings in the Google fact-check index between October 2024 and October 2026. The label is the desk's rating: 211 false, 90 misleading and 19 true. A false rating expects FAKE and accepts PARTLY FALSE or MISLEADING; a misleading rating expects MISLEADING and accepts the other two; a true rating expects REAL. The desk's own ruling is hidden during a run, so the system may use only other desks' rulings on the same claim. We split the set once, before seeing any result: rulings 1–120 for development, where we read failures and corrected the system, and rulings 121–320 sealed, run whole once per release candidate, reported as totals only and logged with the code version whether or not the run completed.

Lookalike publishers. We checked five lookalikes of the BBC, Reuters and the Associated Press, and ran a made-up income-tax story live on one of them.

Providers. Each of ten model tasks ran three times per provider on a Windows 11 laptop with an NVIDIA RTX 5050 (8 GB). Each run was pinned to one provider, so that a failure

could not fall through to another provider and be counted as a success. The hosted models were Nemotron 3 Super 120B on NIM and Gemini 3.1 Flash-Lite.

4 Results

4.1 Held-Out ISOT Articles

Table 3 and Fig. 3 compare the two rules on the 120 test articles. The proposed rule got 114 right (95.0%, 95% interval 89.5–97.7%) and the baseline 115 (95.8%, 90.6–98.2%), so accuracy is the same to within one article. The difference is in calibration on accusations. ECE there fell from 0.267 to 0.152, which is 43% lower, and precision on accusations stayed at 0.967. When the system says a story is probably fabricated, the probability it gives is closer to how often that is true.

Table 3. Held-out comparison of the two scoring rules (120 ISOT articles; lower is better for ECE and Brier).

Rule	Accuracy	ROC-AUC	ECE	ECE (accusing)	Precision (accusing)	Brier
Hand-weighted baseline	0.958	0.978	0.237	0.267	0.968	0.101
Proposed, as deployed	0.950	0.951	0.246	0.152	0.967	0.117

The proposed rule ranks a little worse (ROC-AUC 0.951 against 0.978), and its overall ECE (0.246 against 0.237) and Brier score (0.117 against 0.101) are slightly higher. Both follow from Eq. (3). On ISOT the only evidence that could clear an article is its style, which the rule will not use, so genuine articles stay near the prior. The 59 test articles in the lowest band were given about 0.41 and turned out to be fabricated 7% of the time. In normal use, corroboration and the publisher's record do that job, and neither is available on a historical corpus with the sources hidden. Turning the clamp off raised ROC-AUC to 0.983 and ECE on accusations to 0.191. We kept the clamp.

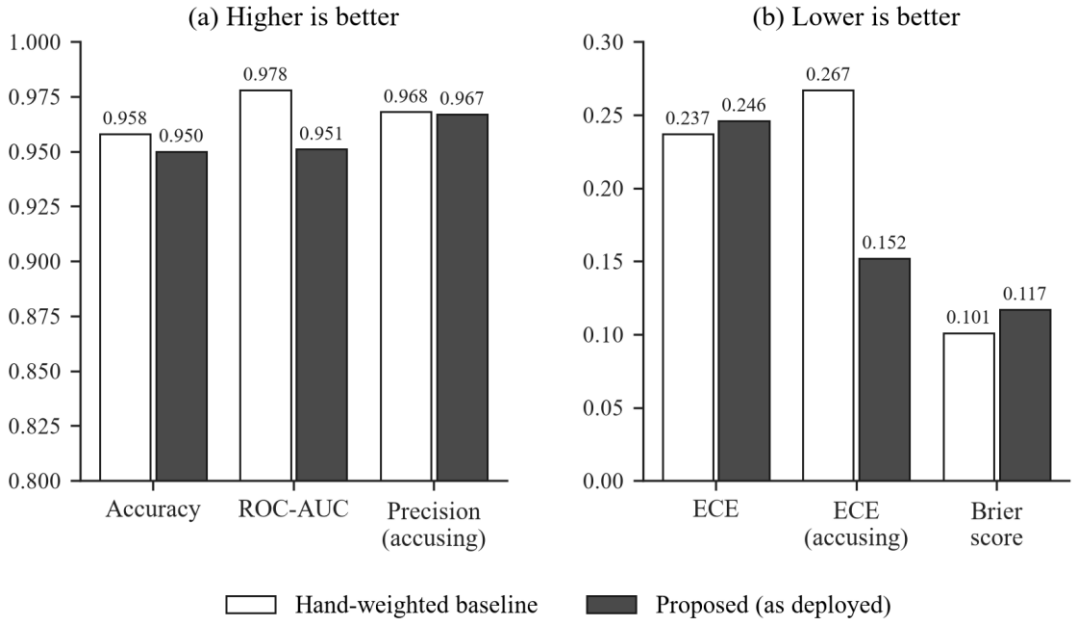


Fig. 3. Held-out metrics for the baseline and the proposed rule: (a) higher is better; (b) lower is better.

4.2 Adversarial Set

The baseline rated four of the six deceptive items as credible. The proposed rule rated none of them as credible, matched the expected verdict on all six and did not condemn either genuine report (Table 4). The fabrications still score well for writing, 7.3 out of 10 on average under the baseline. They just no longer pass on that alone.

Table 4. Adversarial set, scored under both rules from the same outputs of the rule-based analysers. The baseline gives a score, not a verdict.

Measure	Baseline	Proposed
Deceptive items rated credible	4 of 6	0 of 6
Verdicts matching expectation	–	6 of 6
Genuine reports wrongly condemned	–	0 of 2
Mean writing score of the fabrications	7.3 of 10	–

4.3 End-to-End Cases

Table 5 lists the twelve cases. The four genuine news reports were confirmed by three to five independent outlets each, whether they came in as pasted text, a link, a screenshot or Tamil text. The story calling Rahul Gandhi ‘Prime Minister’ was marked FAKE by the premise check, as text and as a screenshot. Fig. 4 shows what the reader sees, including the line from his Wikipedia entry. The two invented stories came back CANNOT VERIFY, and the three inputs with nothing to check were refused. The Associated Press video, a genuine report

whose transcript is mostly an interview in French, came back CANNOT VERIFY because no matching coverage was found on that run. It was not called false. Analyses took 22 to 112 seconds, and refusals were immediate.

The run also hit a real failure. Partway through the first attempt the laptop lost its internet connection, so every news and reference lookup failed. All of the affected cases came back CANNOT VERIFY under the rule ‘verification unavailable’, with the message ‘We could not complete the search, so nothing has been checked’. None was called false. We re-ran those cases when the connection came back, and Table 5 shows the re-run.

4.4 Lookalike Publishers

Table 6 shows the five lookalikes. When the outlet was taken from the name on the page, every one of them inherited the real outlet’s rating. The made-up income-tax story on `bbc-breaking-news.xyz`, which calls itself BBC News, was then treated as an exclusive from a strong outlet and given a 2% probability of fabrication. With the publisher taken from the address, the same story starts from the poor-record prior, gets 33%, and the reader is told the page is posing as the BBC. The real BBC article in our test set is still confirmed.

Table 5. End-to-end cases run on 4 October 2026.

No.	Input	Expected	Result	Rule that decided it	Time
1	Pasted text	REAL	REAL	5 independent outlets	100 s
2	News link	REAL	REAL	3 independent outlets	33 s
3	Screenshot	REAL	REAL	5 independent outlets	112 s
4	Tamil text	REAL	REAL	4 independent outlets	30 s
5	YouTube video	REAL or CANNOT VERIFY	CANNOT VERIFY	No coverage found	55 s
6	Pasted text	FAKE	FAKE	Premise contradicted	22 s
7	Screenshot	FAKE	FAKE	Premise contradicted	30 s
8	Pasted text	CANNOT VERIFY	CANNOT VERIFY	No coverage found	25 s
9	Pasted text	CANNOT VERIFY	CANNOT VERIFY	No coverage found	28 s
10	Pasted text	Refused	Refused	A question	< 1 s
11	Screenshot	Refused	Refused	A call to action	< 1 s
12	News link	Refused	Refused	Site blocks reading	< 1 s

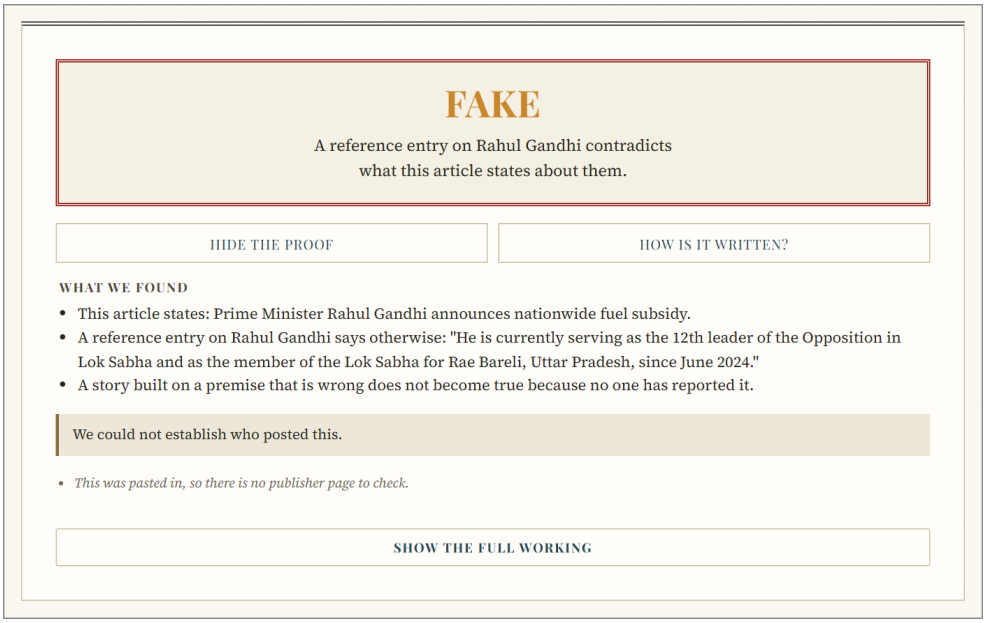


Fig. 4. The report shown for a story built on a false premise (case 6 in Table 5).

Table 6. Lookalike publishers identified by page name and by web address.

Address	By page name	By web address
bbc-breaking-news.xyz, calling itself "BBC News"	BBC News, 9/10	No rating; impersonation flagged
bbc-world.com	BBC News, 9/10	No rating
bbc.news-today.com	BBC News, 9/10	No rating
reuters-updates.info, calling itself "Reuters"	Reuters, 9.5/10	No rating; impersonation flagged
YouTube channel named "Associated Press", not AP's own	AP, 9.5/10	No rating

4.5 Model Routing and Response Time

Fig. 5 shows the median time per call. The local model was faster on every task, by 1.7 to 15.5 times compared with the faster hosted provider. It was also more reliable at claim extraction, where hosted replies were valid in 67% and 33% of runs and local replies in all of them.

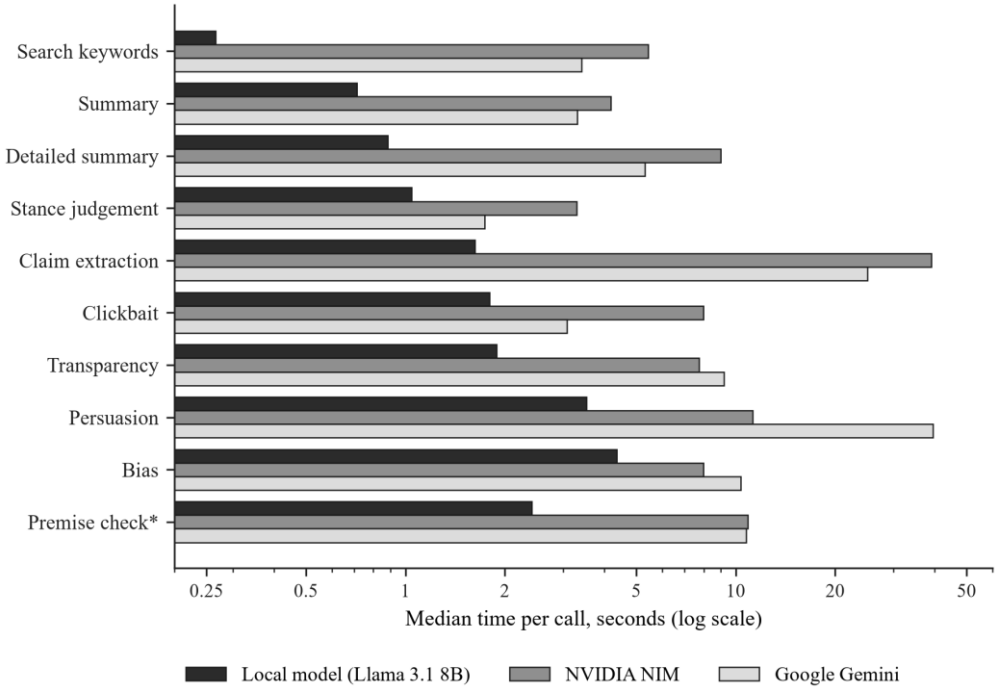


Fig. 5. Median time per call by task and provider (log scale). The premise check (*) is routed to NIM, because the local model missed the known contradiction in all three runs.

The exception was the premise check. Asked whether ‘Prime Minister Rahul Gandhi’ conflicts with his Wikipedia entry, the local model returned clean JSON in 2.4 seconds saying there was no conflict, in all three runs. Both hosted models found the conflict and quoted the sentence that settles it. An earlier version of this benchmark only checked whether replies parsed, and by that measure the local model won this task too, so routing on it would have quietly switched off the premise check. A task with a known answer now has to get it right before it is moved. In the end nine of ten tasks run locally, at no cost per call. In use, the first check reaches the reader 1.47 seconds after an article is opened, and a cached report comes back in 2.7–3.4 milliseconds.

4.6 Messages as People Write Them

Table 7 gives the robustness suite before and after the reading step. Before it, 37 answers were expected, 13 missed and 2 wrong; every failure came from rules alone deciding what a message said. After it, 52 were expected, 1 missed and none wrong. The suite was used during development, so the held-out set is the fairer figure: 24 of 27 expected at first sight. Its worst miss was a true claim carrying an invented one, ‘Modi launched ₹13,000 crore projects in Chennai and also announced free petrol for all college students’, returned as REAL on its first half; rule 11 of Table 2 was added in response. After the fixes, 24 of 27 were expected and 1 wrong.

Table 7. Robustness suite before and after reading the message (expected / missed / wrong).

Kind of message	Cases	Before	After
Plain claims, numbers, typos	7	5 / 2 / 0	7 / 0 / 0
Questions, passed-on claims	7	3 / 4 / 0	7 / 0 / 0
Forwards, sarcasm	4	4 / 0 / 0	4 / 0 / 0
Negation, opinion	5	4 / 0 / 1	4 / 0 / 0
Several claims	2	1 / 0 / 0	2 / 0 / 0
Tamil, Tanglish, Hindi	13	10 / 3 / 0	13 / 0 / 0
Old news, standing facts	7	3 / 3 / 1	7 / 0 / 0
Quotes, local, rumours, links	7	6 / 0 / 0	6 / 1 / 0
Forecasts, vague text	3	1 / 1 / 0	2 / 0 / 0
Total	55	37 / 13 / 2	52 / 1 / 0

4.7 Against a Plain Language Model

Table 8 gives the benchmark. The plain model answered 15 of 22 correctly and the system 22. The plain model called 4 of the 9 true reports fake, among them the Reserve Bank's rate rise and India's 85-medal total, so its 'fake' on a false claim is a habit rather than a finding, and it cites nothing a reader can check.

Table 8. Correct answers on current claims, by kind of claim.

Kind of claim	Claims	Plain model	Our system
Wrong figure	2	2	2
Reversed outcome	2	1	2
Viral rumour	3	3	3
Invented event	4	4	4
False premise	2	2	2
True report	9	3	9
Total	22	15	22

4.8 Fact-Checked Rulings

Table 9 gives the rulings set. The first release candidate gave the desk's verdict for 97 of the 200 sealed rulings, an acceptable verdict for 34, no verdict for 52 and a wrong verdict for 17. Its wrong calls on claims rated true led to one more round of corrections, made on the development rulings and on a second development pool of 160 rulings from outside India, never on sealed cases. In that round we found that our reader of desk ratings had taken ratings

written as sentences, such as ‘That’s not correct’, as true; a script that reads the ratings alone, without the system’s answers, corrected four sealed labels from true to false. The first candidate’s totals stand as recorded, on the old labels, and cannot be rescored because a sealed run keeps totals only, so part of the later fall in wrong verdicts may be due to the labels rather than the code. The final code gave 102, 39, 47 and 12, at a median of 9.3 seconds a claim; a first attempt at that run stopped after 31 rulings when the free model quota ran out, and is logged as incomplete. Claims rated true are the weak point, with 3 of 13 confirmed: in development, such failures came from events too thinly reported for two independent outlets, rankings that no headline states, and counts accumulated over different dates. The result also leans on the hosted model. On rulings 1–60, with Gemini the system gave 27, 10, 21 and 2; without it, on NIM and the local model, 15, 6, 37 and 2, no more wrong calls but many more claims left without a verdict.

Table 9. Fact-checked rulings: desk’s verdict / acceptable / no verdict / wrong.

Rulings and code	Claims	Desk’s	Accept.	None	Wrong
Development 1–120, final code	120	57	27	33	3
Sealed 121–320, first candidate	200	97	34	52	17
Sealed 121–320, final code	200	102	39	47	12
Final, sealed: rated false	130	98	4	26	2
Final, sealed: rated misleading	57	1	35	14	7
Final, sealed: rated true	13	3	0	7	3

4.9 Explanations, Claim Memory and Our Own Model

Every check is kept as an append-only record of the exact inputs the verdict engine read, so a stored verdict can be reached again from its record. Rewordings and translations of a claim are filed under one record, and a changed figure, date, outcome or role is linked as a variant rather than merged. On 46 held-out claim pairs, with our own pair model settling the unclear ones, the matcher filed 44 correctly, merged no two different claims and kept all 18 decoys apart.

Because rules decide the verdict, the reader can be told what decided it and what would change it, and each such statement can be checked. We compute it by running the same rules again with each line of evidence removed, or with hypothetical evidence added. A separate script recomputed every statement on 94 stored checks, 380 scenario ledgers and 2000 random ones, and all 33,429 agreed. Over 20,000 further random ledgers, adding a fact-

checker's false rating, an encyclopedia contradiction or two more outlets reporting otherwise never made a verdict milder, and adding confirming evidence never made it harsher.

Our own pair model is a multilingual MiniLM with 118M parameters. We fine-tuned it on a laptop GPU on English, Tamil and Hindi sentence pairs and on fact-checkers' claim pairs split by rumour, and exported it as a 42 MB int8 model that judges a pair in 8 to 12 ms on the server's processor. On 72 hand-written blind pairs it is right in 76%. It matches claims to fact-checks and to earlier checks, and vetoes contradictions that a fallback model reports but the encyclopedia sentence does not state. Changed details in a true story are caught by comparison rather than by a model: figures and codes in the claim are matched against headlines and snippets, so 'sent to 4 days in judicial custody' is reported false against outlets that say five. Untrusted text is fenced in every prompt and model replies are validated strictly; nine of nine live injection attempts in English, Hindi and Tamil failed.

4.10 Comparison with Published Results

Table 10 puts our numbers next to DEFAME's and next to the ISOT shortcut reported by Verma [4], who found that a classifier given only the subject field of each article, with the text thrown away, reaches F1 = 1.000. The rows use different data, labels and tasks, so this is not a controlled comparison, and we did not run DEFAME on our cases. The table shows where each number comes from, and that a perfect ISOT score is possible without reading a single article.

5 Discussion

5.1 What the ISOT Score Shows

ISOT accuracy says little on its own. Verma [4] showed how much of it comes from the corpus, and our held-out set is small: the 95% intervals of the two rules overlap almost completely. What our ISOT test does show is that the one-sided rule costs very little accuracy on ordinary data while making accusations better calibrated. The adversarial set is the test that matters for this design, and there the gap between the rules is clear, four credible against none.

5.2 Why the Rule Costs Ranking

The drop of 0.032 in ROC-AUC against the unclamped model is the price of Eq. (3). Without the clamp the model learns from ISOT that tidy writing means a genuine story, which is true of that corpus and false against a careful writer. We would rather give up a little ranking on

old data than reward the one thing a fabricator can always supply. On ISOT the largest calibration gap is also on the cautious side, with the lowest band stated at 0.41 and observed at 0.07. Genuine stories are meant to be cleared by corroboration and the publisher's record, which ISOT cannot test.

Table 10. Reported results of related work and of our two rules; data and tasks differ between rows.

Study	Data	Task	Reported result
DEFAME, GPT-4o [1]	AVeriTeC	Real-world claims, web evidence	Acc. 70.5%
DEFAME, GPT-4o [1]	VERITE	Image and text claims, true vs false	Acc. 83.9%
DEFAME, GPT-4o [1]	ClaimReview2024+	Claims after the model's cut-off	Acc. 69.7%
Subject field only [4]	ISOT	Fake vs real, no article text	F1 1.000
Our baseline	ISOT, 120 held out	Fake vs real, source hidden	Acc. 95.8%
Our proposed rule	ISOT, 120 held out	Fake vs real, source hidden	Acc. 95.0%, accusing ECE 0.152
Our proposed rule	Adversarial set	Six careful fabrications	0 of 6 rated credible
Our system	Plain-model benchmark	Claims from October 2026	22 of 22 correct
Gemini, no evidence	Plain-model benchmark	Claims from October 2026	15 of 22 correct
Our system	Fact-checked rulings, 200 sealed	Claims ruled on by desks	141 desk's or acceptable, 12 wrong

5.3 Use in Practice

For a platform or a messaging service, a verdict that comes with its reason, and a willingness to say CANNOT VERIFY, is more useful than a confident score. For newsrooms and other organisations whose name carries weight, such as banks and government departments, reading identity from the address stops copycat sites from borrowing their record. For regulators, every verdict can be traced to a rule, the outlets involved and quoted text, so it can be checked and appealed, and opinion pieces are refused instead of judged. Messages in Tamil and Hindi, or typed in English letters, are read as readily as English, and the one-line reason is given back in the reader's script. Where a machine with a GPU is available, running eight of sixteen model tasks locally also removes per-call charges for those tasks and keeps

the article text on that machine, which matters for low-budget public deployments in India. In this way the work supports SDG 16, target 16.10, on public access to information.

5.4 Threats to Validity

The likelihood ratios come from 239 articles, so some bands are too thin to estimate and a one-article difference in accuracy means nothing. Each result comes from a single run. The adversarial set is small and we wrote it ourselves, so it shows that the rule behaves as designed, not how often it works in the wild. It was scored with the rule-based analysers, and a full live run is still to be done. The publisher priors and the corroboration, ruling and coverage ratios are set by hand. The robustness, held-out and benchmark sets were written by us from one month's news, and their labels hold only while that news is current. The rulings set is labelled by the desks, but only 19 of its 320 claims are rated true, too few to measure that class closely, and its labels pass through our reader of the ratings, which once misread four. Verdicts depend on Google News, the fact-check index, Wikipedia and a hosted model whose free daily quota, once spent, leaves slower fallback models at 60 to 110 seconds a check. Stance is judged from headlines, not full articles. Manipulated images and synthetic video are not analysed directly, though a viral fake video is still caught when fact-checkers have ruled on it.

6 Conclusion

We built a fake news checker that weighs evidence by who controls it. Writing signals can only count against a story, and support has to come from independent outlets, reference entries and a publisher identified by its web address. Every weight is a likelihood ratio estimated from data, and the weights add up to a calibrated probability. On 120 held-out ISOT articles the system matched a hand-weighted baseline's accuracy to within one article and cut calibration error on its accusations by 43%. It rated none of six carefully written fabrications as credible, against four for the baseline, and gave an acceptable verdict in all twelve end-to-end cases. Read for what they assert, messages in three languages and every tone we tried gave 52 expected answers in 55 with none wrong, and on current claims the system answered 22 of 22 correctly against 15 for a language model without evidence. On 200 claims ruled on by fact-checkers and sealed before development, it gave the desk's or an acceptable verdict for 141, no verdict for 47 and a wrong verdict for 12. Next, we will work on the claims it leaves without a verdict and on true claims, which need evidence from the body of an article rather than its headline, grow the estimation corpus so that the priors can be

measured, collect ratings for regional Indian outlets, run the judgements on our own models, and add image forensics as one more observation.

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